

Latex Practice

Simon

May 2026

Question 1

Parameterize S by $\vec{r}(u, v)$ over a region $D \subset \mathbb{R}^2$. Writing $\mathbb{F} = (P, Q, R)$, the line integral becomes

$$\oint_{\partial S} P dx + Q dy + R dz = \oint_{\partial D} \tilde{P} du + \tilde{Q} dv,$$

where $\tilde{P}, \tilde{Q}$ absorb the Jacobian terms from the chain rule. Applying Green's theorem to the right hand side,

$$\iint_D \left(\frac{\partial \tilde{Q}}{\partial u} - \frac{\partial \tilde{P}}{\partial v} \right) du dv.$$

Expanding via the chain rule and collecting terms yields exactly $\nabla \times \mathbf{F} \cdot (\vec{r}_u \times \vec{r}_v)$, which equals $(\nabla \times \mathbf{F} \cdot d\mathbf{S})$.

□

Question 2

Let $k = \dim(\ker T)$ and let $\{u_1, \dots, u_k\}$ be a basis for $\ker T$. Extend this to a basis of V :

$$\mathcal{B} = \{u_1, \dots, u_k, v_1, \dots, v_r\}, \dim V = k + r$$

We claim $\{T(v_1), \dots, T(v_r)\}$ is a basis for $\operatorname{im} T$.

Spanning. For any $w \in \operatorname{im} T$ write $w = T(v)$ with $v = \sum_{i=1}^k \alpha_i u_i + \sum_{j=1}^r \beta_j v_j$. Since $T(u_i) = 0$:

$$w = \sum_{j=1}^r \beta_j T(v_j).$$

Independence. Suppose $\sum_{j=1}^r \beta_j T(v_j) = 0$. Then

$$T \left(\sum_{j=1}^r \beta_j v_j \right) = 0$$

so $\sum_{j=1}^r \beta_j v_j \in \ker T$. Writing this as a combination of $\{u_i\}$ and comparing with $\mathcal{B}$ being a basis forces all $\beta_j = 0$.

Hence $\dim(\text{im } T) = r$ giving

$$\dim(\ker T) + \dim(\text{im } T) = k + r = \dim V.$$

□

Question 3

Form the augmented matrix $[A \mid I]$ and apply elementary row operations $E_1, E_2, \dots, E_k$:

$$[A \mid I_n] \xrightarrow{E_1, \dots, E_k} [\text{rref}(A) \mid E_k \dots E_1].$$

Each E_i is invertible and $\det(E_i) \neq 0$.

If A is invertible then $\det(A) \neq 0$, so $\text{rref}(A) = I_n$ (has no row of zeros). The operations yield

$$\left[I_n \mid E_k \dots E_1 \right],$$

and $E_k \dots E_1 = A^{-1}$ since $E_k \dots E_1 \cdot A = I_n$.

If row reduction reaches I_n , then $E_k \dots E_1 A = I_n$ so $A^{-1} = E_k \dots E_1$ exists and A is invertible.

□

Question 4

Let $Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. We prove the matrix identity by induction on n .

Base case $n = 1$:

$$Q^1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix}.$$

Inductive step. Assume $Q^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$. Then

$$\begin{aligned} Q^{n+1} &= Q^n \cdot Q \\ &= \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} F_{n+1} + F_n & F_{n+1} \\ F_n + F_{n-1} & F_n \end{pmatrix} \\ &= \begin{pmatrix} F_{n+2} & F_{n+1} \\ F_{n+1} & F_n \end{pmatrix}. \end{aligned}$$

This completes the induction.

For Binet's formula, diagonalize $Q = PDP^{-1}$ where $D = \text{diag}(\varphi, \psi)$, so $Q^n = PD^nP^{-1}$. Reading off the $(2, 1)$ entry gives $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$.

□

Question 5

Let $\{q_1, q_2, \dots\} = \mathbb{Q} \cap [a, b]$.

Step 1: Diagonal extraction. Since $(f_n(q_1))$ is bounded, Bolzano–Weierstrass gives a subsequence $(f_{n_k^{(1)}})$ converging at q_1 . Extract a further subsequence converging at q_1 and q_2 , and so on. the diagonal sequence $g_k := f_{n_k^{(k)}}$ satisfies:

$$g_k(q_j) \rightarrow L_j \in \mathbb{R} \quad \text{for every } j \in \mathbb{N}.$$

Step 2: Uniform convergence on $[a, b]$. Let $\varepsilon > 0$. By equicontinuity choose $\delta > 0$ so that $|x - y| < \delta \Rightarrow |g_k(x) - g_k(y)| < \varepsilon/3$ for all k . Cover $[a, b]$ by finitely many δ -balls centered at rationals $q_{j_1}, \dots, q_{j_r}$. Choose K so that for $k, \ell \geq K$ and $i = 1, \dots, r$:

$$|g_k(q_{j_i}) - g_\ell(q_{j_i})| < \frac{\varepsilon}{3}.$$

For any $x \in [a, b]$ pick q_{j_i} with $|x - q_{j_i}| < \delta$:

$$\begin{aligned} |g_k(x) - g_\ell(x)| &\leq |g_k(x) - g_k(q_{j_i})| + |g_k(q_{j_i}) - g_\ell(q_{j_i})| + |g_\ell(q_{j_i}) - g_\ell(x)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

So (g_k) is uniformly Cauchy, hence uniformly convergent.

□

Question 6

Theorem - Leibniz integral rule

Let $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$ satisfy: (i) $F(x) = \int_c^d f(x, t) dt$ exists for all $x \in [a, b]$, (ii) $\partial_x f(x, t)$ exists for all (x, t) , (iii) $|\partial_x f(x, t)| \leq g(t)$ for some $g \in L^1[c, d]$. Then F is differentiable and $F'(x) = \int_c^d \partial_x f(x, t) dt$.

Proof. Fix $x \in [a, b]$. For $h \neq 0$ small, define

$$\varphi_h(t) := \frac{f(x+h, t) - f(x, t)}{h}.$$

Then $\frac{F(x+h) - F(x)}{h} = \int_c^d \varphi_h(t) dt$. By the Mean Value theorem, for each t there exists $\xi_h(t)$ between x and $x+h$ with

$$\varphi_h(t) = \partial_x f(\xi_h(t), t),$$

so by hypothesis (iii),

$$|\varphi_h(t)| = |\partial_x f(\xi_h(t), t)| \leq g(t) \quad \text{for all } h \neq 0 \text{ and } t \in [c, d].$$

Since $\varphi_h(t) \rightarrow \partial_x f(x, t)$ pointwise as $h \rightarrow 0$ and $g \in L^1$ dominates, the Dominated Convergence Theorem gives

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \int_c^d \varphi_h(t) \, dt \\ &= \int_c^d \lim_{h \rightarrow 0} \varphi_h(t) \, dt \\ &= \int_c^d \partial_x f(x, t) \, dt \end{aligned}$$

□